aocseq: Activation of cells and sequencing for the classification of cells in mixed populations

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`AddDistances` *Return a data frame of meta data associatd with single cell data sets that contains a distance from a reference*

Description

This function will compute the distance for each single cell. For each cell in the reference and then return it in a data frame along with all the additional single cell metadata

Usage

```
AddDistances(
  cell.data,
  array.name,
  signature.ref,
  path.glist,
  distance = 0,
  scramble = FALSE,
  file.path = FALSE,
  PRINT = FALSE,
  verbose = TRUE
)
```

Arguments

`array.name`	A data frame containing a list of TCR sequences, the frequency of cells expressing those sequences in the CD4 and CD8 population and total number of cells
`signature.ref`	Directory containing gene set
`path.glist`	Directory of cell type labels stored in csv format
`distance`	Directory of output images
`scramble`	A Seurat object containing the distribution used to generate mahalanobis distance
`file.path`	Distance function, default is 0=Mahalanobis, 1=Taxicab, 2=1D-Mahalanobis
`PRINT`	Distance function, default is 0=Mahalanobis, 1=Taxicab, 2=1D-Mahalanobis
`cell.array`	A Seurat object containing cells that are to be assigned a distance

`AnnotateClonotypes` *List clonotypes and the proportion of cells expressing a choice of marker genes*

Description

This function will read in Seurat objects processed by aocseq::CombineData and generate a ranked spreadsheet of clonotypes grouped by CDR3 beta sequence that displays the percentage of CD4 and CD8 cells with transcript expression that is greater than the quantile cutoff of the control set in aocseq::CombineData. A gene list used to estimate specificity can be chosen and is application specific. Other parameters are listed for debugging, but can be left as default values.

Usage

```
AnnotateClonotypes(
  Clonal_Obs,
  goi,
  goi.threshold,
  index.control = 1,
  conditions = 1,
  path = "",
  names.spreadsheet = 0,
  preset = 1,
  threshold.entry = FALSE
)
```

Arguments

`Clonal_Obs`	A Seurat object pre-processed with traceseq::CombineData.
`goi`	A marker of interest.
`goi.threshold`	List. Percentile used for specifying high versus low expression for each condition.
`index.control`	List. Index of the control sequencing sample for each condition.
`conditions`	Number of experiments within ClonalData object
`names.spreadsheet`	List. Control and sample names.
`preset`	Bool. If set to 1, the control is part of the dataset. If 0, a threshold value for each condition must be set.
`threshold.entry`	List. List of thresholds for each condition if preset is 0.
`goipos`	Index of the goi in the complete marker list.
`clonotype_path`	path for output of the clonotype spreadsheet.
`n.batch`	Number of batches or hashtags for each condition.

Value

A data frame containing clonotypes and summary statistics of transcript expression.

`ClassifyCells` *Return mahalanobis distance of query dataset compared to reference*

Description

This function will compute the mahalonis distance for each single cell. For each cell in the reference, we construct the positive definite covariance matrix of the reference data set signature.ref. Takes as input two Seurat objects, one containing the query cells and another containing the reference. A gene list used to construct the distance and output path for plotting. Other parameters are listed for debugging, but can be left as default values. there are different functions for different genesets

Usage

```
ClassifyCells(
  cellnopf.array,
  cell.array,
  signature.ref,
  path.glist,
  Glist,
  distance = 0,
  scramble = FALSE,
  withSCT = FALSE,
  verbose = TRUE
)
```

Arguments

`cell.array`	A Seurat object containing cells that are to be assigned a distance
`signature.ref`	A Seurat object containing the distribution used to generate mahalanobis distance
`path.glist`	Directory containing gene set
`scramble`	Print maximum distance for dataset neighbors. FALSE by default
`verbose`	Print progress bars and output
`summary.df`	A data frame containing a list of TCR sequences, the frequency of cells expressing those sequences in the CD4 and CD8 population and total number of cells
`path.outs`	Directory of cell type labels stored in csv format
`path.figs`	Directory of output images
`which.gene`	Sets the gene used for ploting
`specificity.percentile`	A cutoff for the mahalanobis distance
`plotting.max`	A maximum value for the speicificty displayed in the plots
`plotting.percentile`	A cutoff for the specificity of the plots

Value

A Seurat object containing metadata with mahalanobis distances.

`ClassifyClonotypes` *Combine single cell data and annotate with cell labels based on functionality*

Description

This function will read in 10X genomics outputs from cellranger and combine multiple Seurat objects, assigning labels based on additional tables stored in csv format. Takes as input gene expression from scRNAseq and VDJ enrichment. A gene list used to estimate specificity can be chosen and is application specific. Other parameters are listed for debugging, but can be left as default values.

Usage

```
ClassifyClonotypes(
  Clonal_Obs,
  Clonotypes,
  goi,
  method = "ZINB",
  path = "",
  hashtags = 0,
  verbose = TRUE
)
```

Arguments

`hashtags`	Integer represents the number of hashtags used (if any)
`verbose`	Print progress bars and output
`gex.path`	path to gene expression data in the format of cellranger feature barcode matrices
`vdj.path`	path to VDJ expression data in the format of cellranger csv files
`marker.gene`	list of marker genes used for specificity analysis
`threshold.cutoff`	list of marker genes used for specificity analysis
`save.dir`	directory for storing Seurat object RDS files
`index.control`	index of the control sequencing sample
`preset`	If set to 1, preset determines the threshold for cutoff if a control is included in the assay
`demultiplex`	Boolean value used to indicate if the data was hashtagged

Value

A Seurat object list containing metadata and VDJ annotations.

`CombineData` *Combine single cell data and annotate with cell labels based on functionality*

Description

This function will read in 10X genomics outputs from cellranger and combine multiple Seurat objects, assigning labels based on additional tables stored in csv format. Takes as input gene expression from scRNAseq and VDJ enrichment. A gene list used to estimate specificity can be chosen and is application specific. Other parameters are listed for debugging, but can be left as default values.

Usage

```
CombineData(
  gex.path,
  vdj.path,
  marker.gene,
  threshold.cutoff,
  save.dir,
  index.control,
  n.samples,
  c.index,
  sample.name,
  preset = 1,
  threshold.entry = 0,
  demultiplex = FALSE,
  n.hashtag.samples = 1,
  n.samples.ht = 1,
  tenX_conversion = "true",
  hashtags = 1,
  nFeature_RNA_lower = 100,
  nFeature_RNA_upper = 10000,
  percent.mt_upper = 5,
  verbose = TRUE
)
```

Arguments

`gex.path`	path to gene expression data in the format of cellranger feature barcode matrices
`vdj.path`	path to VDJ expression data in the format of cellranger csv files
`marker.gene`	list of marker genes used for specificity analysis
`threshold.cutoff`	list of marker genes used for specificity analysis
`save.dir`	directory for storing Seurat object RDS files
`index.control`	index of the control sequencing sample
`preset`	If set to 1, preset determines the threshold for cutoff if a control is included in the assay
`demultiplex`	Boolean value used to indicate if the data was hashtagged

hashtags	Integer represents the number of hashtags used (if any)
verbose	Print progress bars and output

Value

A Seurat object list containing metadata and VDJ annotations.

DEsingle *This is a copy of DEsingle that was lifted and included in the aocseq package*

Description

This is a copy of DEsingle that was lifted and included in the aocseq package

Usage

```
DEsingle(counts, group, parallel = FALSE, BPPARAM = bpparam())
```

Arguments

counts	A non-negative integer matrix of scRNA-seq raw read counts or a `SingleCellExperiment` object which contains the read counts matrix. The rows of the matrix are genes and columns are samples/cells.
group	A vector of factor which specifies the two groups to be compared, corresponding to the columns in the counts matrix.
parallel	If FALSE (default), no parallel computation is used; if TRUE, parallel computation using `BiocParallel`, with argument `BPPARAM`.
BPPARAM	An optional parameter object passed internally to `bplapply` when `parallel=TRUE`. If not specified, `bpparam()` (default) will be used.

Details

DEsingle: Detecting differentially expressed genes from scRNA-seq data

This function is used to detect differentially expressed genes between two specified groups of cells in a raw read counts matrix of single-cell RNA-seq (scRNA-seq) data. It takes a non-negative integer matrix of scRNA-seq raw read counts or a `SingleCellExperiment` object as input. So users should map the reads (obtained from sequencing libraries of the samples) to the corresponding genome and count the reads mapped to each gene according to the gene annotation to get the raw read counts matrix in advance.

Value

A data frame containing the differential expression (DE) analysis results, rows are genes and columns contain the following items:

- theta_1, theta_2, mu_1, mu_2, size_1, size_2, prob_1, prob_2: MLE of the zero-inflated negative binomial distribution's parameters of group 1 and group 2.
- total_mean_1, total_mean_2: Mean of read counts of group 1 and group 2.
- foldChange: total_mean_1/total_mean_2.

- norm_total_mean_1, norm_total_mean_2: Mean of normalized read counts of group 1 and group 2.
- norm_foldChange: norm_total_mean_1/norm_total_mean_2.
- chi2LR1: Chi-square statistic for hypothesis testing of H0.
- pvalue_LR2: P value of hypothesis testing of H20 (Used to determine the type of a DE gene).
- pvalue_LR3: P value of hypothesis testing of H30 (Used to determine the type of a DE gene).
- FDR_LR2: Adjusted P value of pvalue_LR2 using Benjamini & Hochberg's method (Used to determine the type of a DE gene).
- FDR_LR3: Adjusted P value of pvalue_LR3 using Benjamini & Hochberg's method (Used to determine the type of a DE gene).
- pvalue: P value of hypothesis testing of H0 (Used to determine whether a gene is a DE gene).
- pvalue.adj.FDR: Adjusted P value of H0's pvalue using Benjamini & Hochberg's method (Used to determine whether a gene is a DE gene).
- Remark: Record of abnormal program information.

Author(s)

Zhun Miao.

See Also

DEtype, for the classification of differentially expressed genes found by DEsingle.

TestData, a test dataset for DEsingle.

Examples

```
# Load test data for DEsingle
data(TestData)

# Specifying the two groups to be compared
# The sample number in group 1 and group 2 is 50 and 100 respectively
group <- factor(c(rep(1,50), rep(2,100)))

# Detecting the differentially expressed genes
results <- DEsingle(counts = counts, group = group)

# Dividing the differentially expressed genes into 3 categories
results.classified <- DEtype(results = results, threshold = 0.05)
```

GMM_demux *Flexible implementation of the GMM demux algorithm*

Description

Demultiplex arbitrary number of hashtags

Usage

```
GMM_demux(s.name, data, hashtag_index, nameshashtags)
```

Arguments

`s.name`	Name of the sample
`data`	Input data. 10x object with hashtags
`hashtag_index`	list containing positions of hashtags to be demultiplexed
`nameshashtags`	Subsets of the sample for new orig.ident

Value

A Seurat object that is has

`GetGeneSignature` *Export differential gene expression from several different samples and phenotypes*

Description

This function will take a set of annotation spreadsheets and export differential expression between different cell types labelled by their marker genes Currently the phenotypic separation is CD4 and CD8, but the reader is encouraged to generate as many subsets as needed for downstream analysis

Usage

```
GetGeneSignature(Clonal_Obs, clonotype.path, deg.path, goi)
```

Arguments

`object`	The object used to calculate knn
`nn.idx`	k near neighbour indices. A cells x k matrix.
`assay`	Assay used for prediction
`reduction`	Cell embedding of the reduction used for prediction
`dims`	Number of dimensions of cell embedding
`return.assay`	Return an assay or a predicted matrix
`slot`	slot used for prediction
`features`	features used for prediction
`mean.function`	the function used to calculate row mean
`seed`	Sets the random seed to check if the nearest neighbor is query cell
`verbose`	Print progress

Value

return an assay containing predicted expression value in the data slot

GetSpecificCells — *Subsets a SEURAT object based on a list of meta data for the CD4 and CD8 subsets*

Description

This function will read in Seurat objects processed by traceseq::AnnotateClonotypes or any other software based on cell annotation and subset the SEURAT array accordingly

Usage

```
GetSpecificCells(Clonal_Obs, expression, phenotype, tcrseq, goi_v)
```

Arguments

`Clonal_Obs`	A Seurat object
`expression`	A marker of interest.
`phenotype`	Index of the moi in the complete marker list.
`tcrseq`	path for output of the clonotype spreadsheet.
`goi_v`	Gene of interest.

Value

A data frame containing clonotypes and summary statistics of transcript expression.

MakeReference — *Combine single cell data and annotate with cell labels based on functionality*

Description

This function returns a list of reference data normalized by depth and scTransform

Usage

```
MakeReference(
  Act.data,
  path.glist,
  TCR.list,
  sample.return = 3,
  n.inlist = 10,
  verbose = TRUE
)
```

Arguments

`verbose`	Print progress bars and output
`gex.path`	path to gene expression data in the format of cellranger feature barcode matrices
`vdj.path`	path to VDJ expression data in the format of cellranger csv files
`marker.gene`	list of marker genes used for specificity analysis
`threshold.cutoff`	list of marker genes used for specificity analysis
`mask`	listing of genes to be dropped from analysis
`save.dir`	directory for storing Seurat object RDS files
`index.control`	index of the control sequencing sample
`preset`	If set to 1, preset determines the threshold for cutoff if a control is included in the assay
`demultiplex`	Boolean value used to indicate if the data was hashtagged
`hashtags`	Integer represents the number of hashtags used (if any)

Value

A Seurat object list containing metadata and VDJ annotations.

`m_step` *Maximization Step of the EM Algorithm*

Description

Update the Component Parameters

Usage

```
m_step(x, posterior.df)
```

Arguments

`x`	Input data.
`posterior.df`	Posterior probability data.frame.

Value

Named list containing the mean (mu), variance (var), and mixing weights (alpha) for each component.

Index